

D386

Hardware and Operating Systems Essentials

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A student opens a spreadsheet app on a laptop bought during a back-to-school sale. The app is installed on the SSD. Where must its instructions be for the CPU to execute them?

- A. Loaded into RAM (main memory), from which the CPU fetches them.
- B. In the power supply.
- C. In the GPU.
- D. On the SSD, which the CPU reads instruction by instruction.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A video editor's workstation, which has a large custom case, has 32 GB of RAM, a 2 TB SSD, and an 8 TB hard drive. Which component will the CPU read data from fastest?

- A. The 8 TB hard drive, because it is largest.
- B. The 2 TB SSD.
- C. The 32 GB of RAM.
- D. They are all the same speed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which memory level is the smallest and fastest?

- A. L3 cache.
- B. RAM.
- C. CPU registers.
- D. Hard disk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A network tool shows a byte as 11010110 in binary. The tool was installed by a technician on her first day. What is this value in hexadecimal and decimal?

- A. 0x6D = 109
- B. 0xD6 = 8
- C. 0xE6 = 230
- D. 0xD6 = 214

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

What is 0x3F in binary?

- A. 00111111
- B. 11110011
- C. 00110001
- D. 00111110

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A web browser from a well-known company runs each tab in its own process. Inside one tab, a script and the page renderer share data as threads. The browser was updated last Tuesday. If one tab crashes, what is most likely?

- A. The crash is caused by last Tuesday's update.
- B. Only that tab fails; other tabs keep running.
- C. The whole operating system restarts.
- D. All tabs crash, because processes share memory.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An office server with 8 GB of RAM runs apps that together need 14 GB. The disk light stays on constantly, and simple tasks take minutes. The server sits in a well-cooled closet. What is the best fix?

- A. Increase the page file on the same disk.
- B. Add RAM or reduce the running workload.
- C. Install a faster CPU.
- D. Improve cooling in the closet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A photo app, built by a two-person studio, saves an image to disk. How does it reach the storage hardware?

- A. It asks the BIOS at runtime.
- B. It needs no OS help because it is small.
- C. It writes directly to the disk controller.
- D. It makes a system call, and the kernel performs the write in privileged mode.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A clinic's requirements for a new scheduling server include: "The system must load a patient's schedule in under 2 seconds." The clinic's parking lot was recently repaved. What type of requirement is this?

- A. Functional.
- B. A hardware inventory item.
- C. Non-functional (performance).
- D. Security requirement.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which is a functional requirement for the same scheduling system?

- A. The hardware costs under \$10,000.
- B. The system is available 99.9% of the time.
- C. The system supports 200 users at once.
- D. Staff can book, move, and cancel appointments.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A data job is 60% parallelizable. The team, which just moved to a new building, runs it on 4 cores instead of 1. What is the best-case speedup?

- A. About 1.82×
- B. 4×
- C. 2.5×
- D. 2.4×

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A job is 75% parallelizable. With unlimited cores, what is the maximum possible speedup?

- A. 1.33×
- B. 4×
- C. 75×
- D. Unlimited.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A student opens a spreadsheet app on a laptop ~~bought during a back-to-school sale~~. The app is installed on the SSD. Where must its instructions be for the CPU to execute them?

Target word: where running code lives **Decides:** The CPU fetches instructions from main memory.

A	Correct	Programs are copied from storage into memory to run.
B	Adjacent category	Not a storage or processing component.
C	Confusion set	The GPU handles graphics, not general program flow.
D	Inverted operation	Storage is too slow; code is loaded into RAM first.

2. Answer C Concept 2

A video editor's workstation, ~~which has a large custom case~~, has 32 GB of RAM, a 2 TB SSD, and an 8 TB hard drive. Which component will the CPU read data from fastest?

Target word: fastest listed memory **Decides:** RAM sits above SSD and HDD in the hierarchy.

A	Inverted operation	Capacity rises as speed falls.
B	Adjacent category	Fast for storage, but slower than RAM.
C	Correct	RAM is far faster than any storage drive.
D	Absolute language	Speeds differ by orders of magnitude.

3. Answer C Concept 2 · transfer

Which memory level is the smallest and fastest?

Target word: top of the hierarchy **Decides:** Registers are inside the CPU.

A	Adjacent category	Fast, but below registers and L1.
B	Confusion set	Larger and slower than cache.
C	Correct	Tiny and fastest.
D	Inverted operation	Largest and slowest.

4. Answer D Concept 3

A network tool shows a byte as 11010110 in binary. ~~The tool was installed by a technician on her first day.~~ What is this value in hexadecimal and decimal?

Target word: convert binary **Decides:** 1101 = D, 0110 = 6; 128+64+16+4+2 = 214.

A	Inverted operation	Reverses the two nibbles.
B	Wrong stem number	Counts the bits instead of converting.
C	Confusion set	Misreads 1101 as E (14); 1101 is D (13).
D	Correct	Each 4-bit group maps to one hex digit.

5. Answer A Concept 3 · transfer

What is 0x3F in binary?

Target word: hex to binary **Decides:** 3 = 0011, F = 1111.

A	Correct	Four bits per hex digit.
B	Inverted operation	Swaps the two nibbles.
C	Confusion set	Reads F as 1.
D	Dropped element	Drops the lowest bit; F is 1111, not 1110.

6. Answer B Concept 4

A web browser ~~from a well-known company~~ runs each tab in its own process. Inside one tab, a script and the page renderer share data as threads. ~~The browser was updated last Tuesday.~~ If one tab crashes, what is most likely?

Target word: isolation between processes **Decides:** Each tab is its own process with its own memory.

A	Camouflage	Update timing does not decide isolation.
B	Correct	Separate processes are isolated.
C	Absolute language	A user-mode process crash does not take down the OS.
D	Inverted operation	Threads share memory; processes do not.

7. Answer B Concept 5

An office server with 8 GB of RAM runs apps that together need 14 GB. The disk light stays on constantly, and simple tasks take minutes. ~~The server sits in a well-cooled closet.~~ What is the best fix?

Target word: fix for thrashing **Decides:** Active memory exceeds RAM, causing constant paging.

A	Inverted operation	Adds more paging, not less.
B	Correct	Removes the paging pressure.
C	Adjacent category	The CPU is waiting on disk, not busy computing.
D	Camouflage	Heat is not the cause.

8. Answer D Concept 6

A photo app, ~~built by a two-person studio,~~ saves an image to disk. How does it reach the storage hardware?

Target word: path from app to hardware **Decides:** User-mode apps must go through the kernel.

A	Confusion set	Firmware handles startup, not normal app I/O.
B	Absolute language	Every app, regardless of size, goes through the kernel.
C	Inverted operation	User-mode code cannot access hardware directly.
D	Correct	Only the kernel accesses hardware directly.

9. Answer C Concept 7

A clinic's requirements for a new scheduling server include: "The system must load a patient's schedule in under 2 seconds." ~~The clinic's parking lot was recently repaved.~~ What type of requirement is this?

Target word: requirement type **Decides:** A speed target is about how well, not what.

A	Inverted operation	Functional would be 'display the schedule'.
B	Confusion set	Not a requirement type.
C	Correct	Describes quality of service.
D	Adjacent category	Security is non-functional, but this one is performance.

10. Answer D Concept 7 • transfer

Which is a functional requirement for the same scheduling system?

Target word: pick the functional requirement **Decides:** Functional = a behavior or feature.

A	Confusion set	A cost constraint.
B	Inverted operation	Non-functional (availability).
C	Adjacent category	Non-functional (capacity).
D	Correct	Describes what the system does.

11. Answer A Concept 8

A data job is 60% parallelizable. The team, ~~which just moved to a new building,~~ runs it on 4 cores instead of 1. What is the best-case speedup?

Target word: speedup **Decides:** $1 / (0.4 + 0.6/4) = 1 / 0.55$.

A	Correct	$1 / 0.55 \approx 1.82$.
B	Absolute language	Assumes the whole job scales with cores.
C	Dropped element	$1 / 0.4$ — drops the parallel part's time as if it took zero.
D	Inverted operation	Multiplies 0.6×4 instead of applying the formula.

12. Answer B Concept 8 • transfer

A job is 75% parallelizable. With unlimited cores, what is the maximum possible speedup?

Target word: speedup ceiling **Decides:** As n grows, speedup $\rightarrow 1 / (1 - p)$.

A	Inverted operation	$1 / 0.75$ uses the parallel share instead of the serial share.
B	Correct	$1 / 0.25 = 4$.
C	Wrong stem number	Uses the percentage as a multiplier.
D	Absolute language	The serial 25% caps it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Von Neumann architecture and the instruction cycle	Programs and data share memory; the CPU repeats fetch → decode → execute, moving data over a bus.	Thinking programs run directly from disk storage.	Where instructions live while running, or the order of CPU steps.
2	Memory hierarchy	Registers → cache → RAM → SSD → HDD: each level down is larger and cheaper per byte but slower.	Assuming more storage capacity makes a system faster.	A speed-vs-capacity trade-off.
3	Binary, hex, and data units	Each hex digit equals 4 bits; a byte is 8 bits; place values double in binary (128, 64, 32, 16, 8, 4, 2, 1).	Reading hex digits as decimal or mixing bits and bytes.	A value in 0b, 0x, or with Mb vs. MB.
4	Processes vs. threads	A process has its own memory space; threads within one process share that memory, so they are lighter but can interfere with each other.	Thinking threads have separate memory like processes.	Shared data between tasks or isolation between programs.
5	Virtual memory and thrashing	Virtual memory pages RAM to disk; when active pages exceed RAM, the system thrashes — constant paging and very slow performance.	Adding a faster CPU to fix heavy paging.	High disk activity and slow response with RAM fully used.
6	User mode vs. kernel mode	Applications run in user mode and request hardware access through system calls; only the kernel runs in privileged mode.	Believing apps talk to hardware directly.	An app reading a file or sending network data.
7	Functional vs. non-functional requirements	Functional requirements say what the system must do; non-functional say how well (speed, reliability, security, cost, scalability).	Labeling a response-time target as functional.	Numbers about speed, uptime, or capacity vs. a feature.
8	Parallel speedup limits (Amdahl's law)	$\text{Speedup} = 1 / ((1 - p) + p/n)$, where p is the parallel share and n the cores; the serial part caps gains.	Expecting speed to rise in step with core count.	A parallel percentage and a number of cores.

Course scope (public description): Hardware and Operating Systems prepares learners for concepts in software engineering by providing a foundation of understanding in computer architecture, the history of computing architectures, and operating systems. Additional topics covered include hardware and software stacks and how to choose appropriate hardware and software solutions to meet both functional and non- functional business requirements.